Informativity of sentence information structure: the role of word order

Anita Peti-Stantić

Abstract

Halliday's (1967) introduction of the term information structure virtually initiated the research on semantic relationships between sentence structure elements based on the discursive position of an utterance. His work has greatly contributed to the defining of information structure as a formal element in the pragmatic structuring of propositions in the discourse (Lambrecht 1994) in terms of the relationships between linguistic tools and circumstances on the one hand and/or speaker's intentions on the other.

Given that the analysis of this relationship is almost entirely novel to Slavic linguistics, I intend to present some new data on the clitic and clitic cluster positioning in the linear order of Croatian sentences. I will offer an account of the basic features of the multi-leveled approach to the analysis of examples containing clitics and clitic clusters, which will allow for the establishment of criteria to classify sentences as unmarked or marked from the point of view of information structure. The premise underlying the work reported here is that information structure is part of the conceptual structure. Within the Theory of Parallel Architecture (Jackendoff 2002), conceptual structure is the component of grammar that enters into a systematic relationship with the units of the phonological and syntactic component, determining the informational potential of the sentence.

The empirical part presents preliminary research results on the relationship between the placement of clitics and clitic clusters and the information structure in Croatian, a so-called "free word-order" language. The goal of this study is to set the ground for establishing the constraints to the combinatorial properties of clitic and clitic cluster units and the hierarchy among them.

1. Introduction

Word order and linearization are two major areas of linguistic inquiry within all linguistic traditions: descriptive, theoretical, and typological. Since human languages differ significantly in the flexibility (or the so called freedom) of word-order permutations they allow, and since Croatian belongs to the group of free word-order languages with a very rich repertoire of prosodic as well as morphological and syntactic features, I will be dealing with only one facet of the word-order phenomenon in Croatian, namely clitics and clitic clusters (Cl and ClCl).

Considering that word order is one of the primary factors in the organization of information structure (IS), the linearization process should be seen as a multi-level setting that takes into account at least phonological (prosodic), syntactic and semantic factors. To be able to analyze linearization from this perspective and so as to establish the criteria for labeling a sentence as unmarked or marked from the point of view of IS, one has to assume that IS is an interface grammatical component that is set up as a potential which is realized in concrete utterances. Although it might seem self-evident that markedness would presuppose a base or prototypical position which serves as the starting point, I will start from a somewhat different perspective.

The notion of basic or default order is fairly well defined in the literature. It refers to the order of major clausal constituents, which are characterized in terms of the relative order of the subject, object and verb. For example, the *Encyclopedia of language and linguistics* (2006, 643) defines the basic order in the following way:

> Within the context of typological studies the term basic order, at the sentence level, is typically identified with the order that occurs in stylistically neutral, independent, indicative clauses with full noun phrase (NP) participants, where the subject is definite, agentive, and human, the object is a definite semantic patient, and the verb represents an action, not a state or an event.

It is instantly obvious that this definition is in harmony with various psycholinguistic accounts where authors declare something along the following lines (see Kaiser, Trueswell 2004, 114): "On an intuitive level, native speakers often feel that one of the orders is the basic, 'default' order, while the other orders are perceived to be somehow more unusual."

However, sentences qualified as exhibiting the typologically defined basic order need not be the most frequent ones, and, what is more important, the notion of basic order should not be equated with the most frequent order. Therefore, as many authors acknowledge (Tomlin 1986; Comrie 1981), it is crucial to note that the 'basic' or 'default' order, defined as above, may, but need not, correspond to the statistically dominant word order in a language.

Suffice it to say that the flexibility of linearization in many languages refers not only to the constituent order, but also to the order of highly frequent, but somewhat peculiar units such as clitics (Cl) and clitic clusters (ClCl). In this respect it is vital to address the significant difference between, for example, English and Croatian. While in English already the change in the constituent order reflects the change in the grammatical relations between constituents, this is not the case in Croatian. As is well known, English sentences such as *John saw Mary* and *Mary saw John* reflect the difference in thematic roles (i.e. who does what to whom). Furthermore, certain sentence types do not allow permutations of constituents in such a way, because they results in ungrammatical sentences, as in the following examples: *John ate an apple* and **An apple ate John*, the ungrammaticality being the consequence of semantic, and not syntactic violations.

In Croatian, which is a language with a rich morphology, however, none of this presents a problem, because thematic roles remain the same, and so does the (basic) information these sentences convey. This is enabled by morphological (case)-markings of the arguments, which allow for the change of the constituent order. For instance, the nominative-accusative case markings remain the same in *Ivan je vidio Mariju* (Ivan-N saw Mary-Acc; 'Ivan saw Mary') and *Mariju je vidio Ivan* (Mary-Acc saw Ivan-N; 'Mary was the one whom Ivan saw'). This pattern also extends to inanimate objects: *Ivan je pojeo jabuku* (Ivan-N ate apple-Acc; 'Ivan ate an apple') and *Jabuku je pojeo Ivan* (Apple-Acc ate Ivan-N; 'It was an apple that Ivan ate').

However, although word-order flexibility in Croatian is much greater and it might seem that the situation therefore may be simpler (because everything, or nearly everything is allowed), this is far from the full picture. For the reasons already given, and having in mind the complexity of possible word-order permutations in the so-called free word-order languages, I take the IS to be an axiomatic component of every sentence and I consider the 'basic' word order that order which can be felicitously uttered in the absence of any substantial context ('minimal context' in Fauconnier's (1994) terms), the usual test being whether this order can be employed in response to questions such as "What happened?"[1]

It goes without saying that word order variations are not uniformed, which means that they are not performed without a good (semantic) reason. Hence I consider the quest for answers to be relying on the analysis of interface rules that interconnect the phonological, syntactic and conceptual tier of the grammatical structure.

Information structure: Conceptual and terminological issues

Halliday's (1967) introduction of the term information structure virtually initiated the research of semantic relationships of elements of the sentence structure based on the discursive position of an utterance. Halliday (1967, 200) states that

> any text in spoken English is organized into what may be called 'information units'. The distribution of the discourse into information units is obligatory in the sense that the text must consist of a sequence of such units. But it is optional in the sense that the speaker is free to decide where each information unit begins and ends, and how it is organized internally; this is not determined for him by the constituent structure.

Information unit, depending on whether it is unmarked or marked, might be a clause, less than a clause, more than a clause, or any combination of these. In his discussion, Halliday deals only with "those options where the clause is orga-

---

1 I am aware of the restriction that not all sentences answer this question, which is the case with stative sentences, e.g. *Knjiga leži na stolu* 'The book is lying on the table', *Ivan voli Mariju* 'Ivan loves Marija' etc.

nized into a whole number of (one or more) information units, [because] this makes it possible to consider the distribution of information within the general framework of thematic systems, taking the clause as point of origin" (Halliday 1967, 2001).

Halliday's research has greatly contributed to the defining of information structure as a formal term of pragmatic structuring of proposition in the discourse (Lambrecht 1994), i.e. the relationships between linguistic tools and circumstances and/or speaker's intentions. Based on these assumptions, in contemporary linguistics we mainly speak of a threefold manner in which words take part in sentence information structure – by introducing their lexical meaning, satisfying syntax in various ways and taking part in the word order. The traditional partition and the hierarchy of grammatical representations (phonological, morphological, syntactic and semantic) do not account for the fact that utterances are organized into information units which are not defined by the constituent structure of the sentence but rather vice versa – the relative order of the information structure units defines the hierarchy of interfaces among the components of grammar.

In what follows I will concentrate on word order and account for the variations present in the contemporary Croatian data which allow Cl and ClCl to be placed second within the syntactic or phonological constituent or a clause, as well as to be placed third in the clause.

Acceptability judgments and experimental design

Although some authors distinguish grammaticality and acceptability judgments (grammaticality associated with the linguistic stimulus and acceptability associated with the stimulus as perceived by the speaker), I adopt the view that there is no polarity between the language system and the cognitive system, the consequence being that grammaticality is mirrored in acceptability judgments.

Jackendoff also, among others, does not distinguish grammaticality and acceptability. In his model of grammar, well-formedness is established through a constraint-based formalism while constraints may be violable, so that structural complexity (and less than perfect grammaticality) can arise through constraint conflict (Jackendoff 2007). This theoretical approach rests on the assumption that grammar contains multiple independent sources of generativity, among which phonology, syntax, and semantics are the most prominent ones. The correlations among these structures are established by *interface rules*, which relate substructures generated in different components. Or, in other words, "a phrase or a sentence is well-formed if it has well-formed structures in all components, related in well-formed fashion by all relevant interface rules" (Jackendoff 2007, 358).

From this point of view, data disputes, mainly cases in which grammaticality judgments fail to provide a clear-cut division between fully acceptable and fully unacceptable sentences, signal much more than a "data problem". This has been especially clear in contemporary linguistics. While linguists in the past supported their theoretical research mostly using their own intuitions, today they can underpin their theoretical research with experimental data. In many areas of linguistic inquiry there has already been a substantial body of evidence for the fact that relevant linguistic examples receive varying degrees of acceptability. This phenomenon has been labeled 'gradience'.[2]

If grammaticality and acceptability are taken as one and the same, then grammaticality has to be understood as in a way gradient, which arises from the structural complexity and the hierarchy of possible violations of constraints on the phonological, morphosyntactic and conceptual tier. The gradience of grammaticality judgments, as seen in many experiments, accounts for all these intralinguistic factors, supplementing them with the frequency of usage, conformity to a prescriptive norm, degree of plausibility, etc.

Being fully aware that the main goals of contemporary linguistics are the necessary expansion of the experimental base and the need to resolve data disputes in an appropriate manner (more on that in Sorace and Keller 2005, 1497–1524), in what follows I will present new data on word order in contemporary Croatian with special reference to Cl and ClCl. Clitics, as well as the whole range of word-order phenomena connected with them, are certainly among the most interesting interface phenomena in the Slavic world. Therefore I will present experimental findings and analyze the gradient data on clitic positioning from the perspective of their information structure.

The practice of using introspective judgments as the main – and sometimes the sole – source of empirical validation of linguistic theories has been under attack for years. The charge was initiated by two books (Schuetze 1996 and Cowart 1997) and an article (Bard et al. 1996). The main argument in all these publications was that the acceptability judgments as used by theoretical linguists lack the degree of experimental control that is assumed to be standard in the social sciences. Since they represent an empirical basis for linguistic theory, this method of collecting the data without following procedures and without standard experimental control techniques should be unacceptable. Alongside the criticism

2 I cannot go into detail here, but for more information on gradience in language see Keller (2000), Jackendoff (2002), Newmayer (2003), Sorace and Keller (2005), Jackendoff (2007), McClelland and Bybee (2007), Wasow (2009).

and the demand for the use of replicable, falsifiable and procedurally strict methods of acceptability judgments, it has been noted that the use of these judgments should not stop at merely distinguishing between acceptable versus unacceptable utterances.

Hence, accounting for the fact that every acceptability/grammaticality judgment represents a sort of metalinguistic performance, one has to find the most suitable method to collect data on a specific phenomenon. I decided to test both the classical Likert-scale method and the magnitude estimation (ME) technique as its methodological alternative. Both techniques serve to collect acceptability judgments on the grammaticality of the sentence in an appropriate and replicable manner.

Magnitude estimation, as described in several works (Bard et al. 1996; Cowart 1997 and Sprouse 2008 among others) is a technique adapted from psychophysics for getting at acceptability judgments. The use of this technique consists of a modulus (an initial sentence) that has been given a value (a number, percentage and/or a line drawing). Every subsequent stimulus should be related to this model value and compared to it. The basic idea behind the use of this technique is that the subjects should be provided with as many options as possible so that they themselves determine the level of acceptability without the limitations of a 1–5 or 1–7 scale.

In ME, the degree of individual variability is claimed to be much higher than in the standard n-point judgments. Whereas it is difficult to imagine the capacity to distinguish the difference between 74% and 77% in linguistic judgments, it is claimed that a great number of subjects allows for fine-grained results. Although this has worked in psychophysics (with loudness or brightness), ME judgments in linguistics should not be taken for granted. In what follows, I will present the results which will shed additional light on this technique.

Word order and clitics

As suggested in the vast body of literature concerning the place Cl and ClCl occupy in sentence linearization processes, Croatian has been claimed to be the classical Wackernagel-type language. According to what has been claimed in the literature on that matter (see exhaustive bibliographies in Franks and Holloway King 2000 and Anderson 2006), this means that Cl and ClCl exclusively occupy the second position of the sentence and that the only question that remains is how to theoretically account for the different realizations of this second position.

Nevertheless, contrary to common assumptions about strict restrictions in the distribution of Cl and ClCl in contemporary Croatian, I identified several patterns that contradict expected clitic positions. Therefore, I will present a number of examples that disprove the assumption that in contemporary Croatian there are at least two non-violable constraints of Cl and ClCl placement:

1. Cl and ClCl must be placed second in a sentence;
2. Cl and ClCl split constituents so that what is left is also a constituent.[3]

Patterns that do not violate these constraints should be evaluated as canonical or prototypical Cl and ClCl placement patterns in Croatian. However, since Croatian has to add a lot to the diversity of word-order phenomena, researchers noted that the first constraint presents a problem as to how one should interpret the second position, and what immediately comes to mind is whether there is a hierarchy built into this sequence so that splitting can occur only when the clitic is placed second in the clause, or in other positions too.

I will present two patterns that significantly diverge from the expected positions. The first pattern consists of sentences that seemingly violate the specified basic constraints, but are nevertheless rated as acceptable. The second type consists of sentences that violate none of the basic constraints, but are rated as unacceptable or as less acceptable than the ones that violate one of the stipulated constraints. An experimentally attested finding of this sort should help us arrive at a conclusion that setting strict mechanical rules of Cl and ClCl placement misses the point, in that the information structure of the sentence (and not just phonological and syntactic structure) is at play in word-order phenomena.

2. The Study

Aims and hypotheses

The aim of this study is to provide relevant data and to analyze the information structure of sentence types with Cl and clitic ClCl in contemporary Croatian. According to what has been reported in the linguistic literature already mentioned, as well as in standard Croatian, mostly normative grammars, Cl and ClCl should be placed in the so-called Wackernagel's position. For that reason, given both the linguistic and normative account, one would expect sentences with the second placement of Cl and ClCl to be rated highest, and sentences with the placement of Cl and ClCl outside the second position much lower. However, contrary to what has been reported in the literature on Cl and ClCl, the judgment

3 Further restrictions are given in Bošković 2009, although corpus data strongly contradict the arguments given there.

task revealed certain interesting patterns among examples that are supposed to be marginally acceptable and unacceptable.[4]

For any number of reason I assume that the analysis of the word-order phenomena, and especially Cl and ClCl placement, calls for the introduction of an approach which accounts for the interdependence of phonological, syntactic and conceptual domains as proposed in Jackendoff's Parallel Architecture.[5]

Sample and procedure

I conducted an experiment with three sets of participants (students in two high school classes and in one elementary school class).[6] All students were presented with the same set of stimuli. One group of high school students had to decide on their grammaticality/acceptability on a 7-point scale, while the other group had to decide by using the ME technique.[7] The elementary school students were asked to decide on a 7-point scale. There were 40 sentences among which there were 9 fillers which were presented to 76 participants.

As already noted, acceptability judgments are inherently ambiguous because judging on the acceptability of sentences might involve different factors such as syntactic well-formedness and the on-line processing difficulty encountered when parsing a sentence, as well as other factors (Chomsky & Miller 1963, Miller & Chomsky 1963, Schütze 1996, Cowart 1997, Staum Casasanto et al. 2010, Hofmeister et al. 2012).

---

4 It should be assumed that the reason for such a discrepancy between what was reported in the literature and what these preliminary experimental results can tell is a consequence of the fact that most literature on Cl and ClCl in South Slavic languages, especially within the generative framework, relied on the intuitive grammaticality judgments of individual native speakers. However, the experiment conducted for this study was designed so as to control for multiple factors, amongst which various phonological, syntactic and semantic peculiarities were found to be at play.

5 To verify the type of an experiment I conducted, being all written, one has to bear in mind that the prosody is not always overt and that the silent reading and implicit prosody should be considered to be relevant, if not decisive factors in clitic placement. Silent reading has been attested in numerous contemporary psycholinguistic findings (cf. Fodor 2002).

6 I am grateful to Ana Bedek and Mirna Tomašek for their help in conducting the experiments.

7 Since I am not familiar with any experiments that used ME for Croatian, my aim was to test this technique in order to determine whether the results differ in any relevant way from the classical 7-point scale.

Results

The starting point for the study was the Wackernagel's position, defined as the second position in the sentence. From what we know this position could be understood phonologically – after the first accented word, or syntactically – after the first syntactic constituent. It is worth noting that these two notions, phonological second position and syntactic second position, sometimes overlap, but are very often realized as distinct positions.

Sentence types analyzed in this study mostly fall under two general categories, clitic second and clitic third, with one exception, which is listed as a special case:

1. Clitic second
   1.1. Clitic second after the first syntactic and prosodic constituent that consists of a single word (N); sentence type: Sestra *će me* dočekati ispred škole. [8]
   1.2. Clitic second after the first syntactic constituent that consists of an NP that consists of words or two NPs; sentence types: Moja sestra *će me* dočekati ispred škole and Moja sestra i njezina prijateljica *će me* dočekati ispred škole.
   1.3. Clitic second splitting the first syntactic constituent that consists of two words or a complex syntactic constituent consisting of two NPs; sentence types: Moja *će me* sestra dočekati ispred škole and Moja *će me* sestra i njezina prijateljica dočekati ispred škole.[9]
2. Clitic third
   2.1. Clitic third after the first syntactic constituent (NP) followed by a verb; types of sentences: Sestra dočekat *će me* ispred škole, Moja sestra dočekat *će me* ispred škole, Moja sestra i njezina prijateljica dočekat *će me* ispred škole.
3. The special case
   3.1. Clitic splitting the first complex syntactic constituent in such a way that none of the constraints are satisfied, sentence type: Moja sestra i njezina *će me* prijateljica dočekati ispred škole.

When reporting on the results, I used certain simplification in notation. The first reported number stands for the mean value of high school students who rated sentences on a Likert scale from 1 to 7. The second number stands for the mean value of high school students who rated sentences according to the ME methods

8 Cl and ClCl are italicized in sentences. Translations and glosses are given bellow where I analyze the data.

9 There is a contrastive alternative to these sentences, which will also be analyzed.

on a scale 0%–100% which I, for the sake of comparability, reduced to a 0–10 scale.[10] This results in a difference between these two scales. The third number represents the mean value of elementary school students who also rated sentences on a Likert scale from 1 to 7.

In what follows I will present the analysis of several examples which give evidence of provocative and unnoticed details that affect judgment ratings, and, as a consequence to that, the theory of the information structure in free word-order languages.

Clitic second

1.1. Clitic second after the first prosodic and syntactic constituent that consists of one word is the only instance representing the perfect match between prosody and syntax, because this position satisfies both prosodic and syntactic constraints. The type of a sentence presented to the subjects was the following one:

(1) Sestra *će* *me*
sister-N will-3-AUX-Cl me-Acc-Cl
dočekati ispred škole.
wait-Inf in front of school-Acc
'My sister will wait for me in front of the school.'

The ClCl *će me* is placed immediately after the noun *sestra* 'sister', which is the first prosodic constituent and at the same time the first syntactic constituent, NP that consists of a bare N.

Although I controlled for several constraints in the sentence design (such as the length of the constituent, the number of syllables and the type of the syntactic construction in complex NPs), I was not aware that the details in sentence design unrelated to the placement of Cl and ClCl can affect the ratings in a statistically significant manner. Such a difference was never reported in the literature and, to the best of my knowledge, has never been tested for Croatian. However, two variants of this pattern presented in the questionnaire showed a statis-

10 The results of comparative testing with a 7-point scale and ME are in line with the reserve voiced by some researchers concerning the availability and relevance of ME for linguistic testing. This point emerges from my data and experimental design. Although ME is, as I already said, claimed to be a fine-grained technique that provides much more reliable results than any other method, and although mean values do not show such a significant difference, there is a striking discrepancy between variances in experiments that used ME and the 7-point scale. I will report on these details in a separate study.

tically significant difference in the rating. Sentence (2) was rated significantly higher (6.9/9.9/5.9) than sentence (3), which was rated 4.1/4.8/1.9.

The sentences presented to the subjects were:

(2) On *će* *te* otpratiti
he will-3-AUX-Cl you-Acc-PRON-Cl escort-Inf
u kazalište.
in theatre-Acc
‘He will escort you to the theatre.’

(3) Sestru *sam* *mu*
sister-Acc be-AUX-Pres-1-Cl him-PRON-Dat-Cl
vidjela jučer.
saw yesterday
‘I saw his sister yesterday.’

In terms of the prosodic or syntactic constituency there is no noticeable difference between these two sentences (in both sentences the first constituent is at the same time the first prosodic, as well as the first syntactic constituent). The only difference between them is the type of syntactic constituent, where *on* ‘he’ is in the subject position and *sestru* ‘sister-Acc’ is in the direct object position.

In a free word-order language such as Croatian, there is no restriction for the object to precede the subject and one would not expect for these two sentences to be rated differently, although they were.

As we can see, the statistically significant difference between examples (2) and (3) appears across subjects, as well as between elementary school students and high school students. The explanation for the across-subject difference can be found in the fact that in example (3) the direct object was placed initially, while in example (2) it was the pronominal subject that was placed initially. Direct object fronting can be seen as an element that calls for an IS explanation. Being fronted, the direct object is focused because the unmarked word order in Croatian is SVO. From the perspective of language learning and the complexity of the structure, it is especially striking that elementary school children rated sentence (3) so low. One possible explanation would be that they did not adopt scrambling processes, but for any conclusion in this direction a detailed and carefully planned experiment would be necessary. Another explanation might be that they have a different way of conceptualizing what was asked in the task, which may, in fact, go hand in hand with language acquisition. However, the available limited data do not allow for the conclusion as to what led to such a significant difference in ratings.

1.2. The second subtype in this section is clitic second after the first syntactic constituent that consists of an NP that consists of two words or two NPs. This is a subtype that shows no special features. The only interesting point is that for the placement of Cl or ClCl in this position it does not matter how long or how complex the first constituent is, because there is no significant variation in the ratings. Sentences such as Moja sestra *će me* dočekati ispred škole, as well as sentences such as Moja sestra i njezina prijateljica *će me* dočekati ispred škole both satisfy the syntactic constraint, although they do not seem to satisfy the prosodic constraint.

Because of the fact that this subtype was not expected to be particular in any respect, I included only one example of each pattern in the questionnaire. The results show that all subjects rated the sentences belonging to this subtype as perfectly acceptable, although they are labeled as substandard in Croatian grammar. Obviously, the length of the first syntactic constituent does not play any role in the rating effects.[11]

The sentences presented to the subjects were:

(4)	Najbolja	prijateljica	*me*		pozvala
	best	friend-N	me-PRON-Acc-Cl		invited
na	rođendan.				
on	birthday-Acc				

6.5/9.3./6.6.

‘My best friend invited me to her birthday party.’

(5)	Leonov	tata	i	njegov	brat
	Leon’s	dad-N	and	his	brother-N
će		*nas*		odvesti	kući.
will-3-AUX-Cl		us-PRON-Acc-Cl		drive-Inf	home

6.1./8.2./6.7.

‘Leon’s dad and his brother will drive us home.’

1.3. The third subtype in this section is Cl and ClCl splitting the first syntactic constituent $[_{NP}$ [Adj*, N]][12] or a complex syntactic constituent that consists of two NPs $[_{NP}$ [Adj*, N] Conj [Adj*, N]].

11 Apart from this questionnaire, I tested examples with even longer and more complex first syntactic constituents among my colleagues at the University and all of them received almost perfect rating.

12 The asterisk stands for any number of adjectives.

Although in this subtype Cl and ClCl split syntactic constituents, they should still be treated as part of the clitic second pattern, because they satisfy the prosodic constraint by following the first prosodic word. There are three subsets of this subtype. The first subtype is the sentence in which the first syntactic constituent consists of an NP that can further be split into Adj+N, the second subtype is the sentence in which the initial NP is split into Adj+Adj+N, and the third subtype is the special case analyzed separately:

(6)	Ivanov	*će*	*me*	stariji
	Ivan's	will-3-AUX-Cl	me-Acc-PRON-Cl	older

brat	sutra	voditi	u	kino.
brother-N	tomorrow	take-Inf	in	cinema-Acc

5.7./7.2./5.9.

'Ivan's older brother will take me to the cinema tomorrow.'

As expected, sentence (6) received a high rating.[13] Further investigations in this direction should be pursued so as to determine whether there is any difference in the ratings for different subsets of examples such as NPs consisting of only an Adj and a N [$_{NP}$ [Adj*, N]], of two nouns [$_{NP}$ [(Adj*), (Poss Pron), N, N]] (such as "prijateljica Marija", '(my) friend Marija'), of possessive pronouns, adjectives and nouns [$_{NP}$ [Poss Pron, Adj*, N]] (such as "moja prijateljica Marija" 'my friend Marija'), or of personal names (name and surname) as a special pattern, as well as other subtypes.

To put the data in line with the gradient perspective, the next example should be the one where the first syntactic constituent consists of a coordinated NP structure of the following type [$_{NP}$ [N] Conj [Poss Pron, Adj*, N]]. In accordance with syntactic accounts of constituency, this example should be rated better than the following one, where the first syntactic NP consists of two coordinate NPs, each of them consisting of at least Adj+N. However, this is not the case.

(7)	Vito	*će*	*me*	i
	Vito-N	will-3-AUX-Cl	me-PRON-Acc-Cl	and

njegova	sestra	dočekati	na	kolodvoru.
his	sister-N	wait-Inf	on	station-L

3.0/2.6/3.3

13 There are numerous examples of sentences that belong to this type in everyday press, as well as in the electronic corpuses of Croatian language.

‘Vito and his sister will wait for me at the station.’

Adding an explicit contrastive reading to the sentence in question does not help its rating, which is also an unexpected result. The only group that rated the sentence with an explicit contrastive reading slightly higher (but not statistically significant) than the sentence without contrastive reading, were high school students who were tested with the use of the ME methodology.

(8)	Vito	*će*		*me*		i
	Vito-N	will-3-AUX-Cl		me-PRON-Acc-Cl		and
njegova	sestra,	a	ne	Juraj	i	njegova
his	sister-N	and	not	Juraj-N	and	his
sestra,	dočekati	na	kolodvoru			
sister-N	wait-Inf	on	station-L			

2.9./3.1./2.6.

‘Vito and his sister, and not Juraj and his sister, will wait for me at the station.’

The next stage within this most complex subgroup are the sentences in which the first syntactic constituent consists of two coordinate NPs, each consisting of NPs that consist at least of an adjective and a noun, the structure represented as $[_{NP}$ [Poss Pron, Adj*, N] Conj [Poss Pron, Adj*, N]].

The example presented to the subjects was:

(9)	Ivanov	*će*	*me*		
	Ivan’s	will-3-AUX-Cl	me-PRON-Acc-Cl		
brat	i	njegov	prijatelj	s	tenisa
brother-N	and	his	friend-N	from	tennis-G
odvesti	na	more.			
drive-Inf	on	sea-Acc			

3.4./4.3./4.9.

‘Ivan’s brother and his friend from tennis will drive me to the coast.’

As can be seen, this sentence was rated significantly higher than the sentence with a single noun in the sentence initial position within the coordinated first syntactic constituent.

A somewhat special case of this subtype is Cl and ClCl second which splits the complex syntactic constituent NP that consist of two complex NPs so that the Cl and ClCl is placed after the first complex NP. Such an example lies somewhere in between patterns 1.2 and 1.3, because this is the example in which Cl and ClCl follow the first syntactic constituent that consists of two lexical and prosodic words, but also splits the first complex syntactic constituent that consists of two NPs. A pattern of this kind is:[14]

(10)	Moja sestra		*će*	*me*	
	My sister-N		will-AUX-Cl	me-PRON-Acc-Cl	
i	njezina	prijateljica	dočekati	ispred	škole.
and	her	friend-N	wait-Inf	in front of	school-G

‘My sister and her friend will wait for me in front of the school.’

It remains to be seen how native speakers would judge this type of sentence, alongside with others that arose in the course of this research.

Clitic third

Clitic third is one of the most intriguing WO patterns because of frequent claims that Croatian is a rigid Wackernagel type of language in which Cl and ClCl are placed sentence second. The second position is also favored in traditional grammars of Croatian (which are for the most part descriptive and to a certain extent normative).[15]

I treat clitic third to be the position following two syntactic constituents (usually NP and VP). Defined in such a simple manner, clitic third pattern splits into three different subtypes, because the first syntactic constituent, as has been shown before, can consist of a simple NP (only N), but it can also consist of a complex NP [$_{NP}$ [Poss Pron, Adj*, N]] or [$_{NP}$ [Poss Pron, Adj*, N] Conj [Poss Pron, Adj*, N]], where we can keep adding more than one adjective and more than one NP. Types of sentences for these cases are: Sestra dočekat *će me* ispred škole. Moja sestra dočekat *će me* ispred škole. Moja (jako pametna starija) sestra i njezina (isto tako pametna i jako zanimljiva) prijateljica (iz srednje škole) dočekat *će me* ispred škole.[16]

14 I did not include this type of sentence in my questionnaire, so I do not have data on acceptability ratings.

15 In one of the most influential grammars (Katičić 1986, 495) it has been claimed that placing Cl and ClCl after the first complex NP, and especially placing clitics in the third position is “a characteristic of substandard spoken expression”. Other grammars do not give similar qualifications.

16 Translation for the last one: My (very clever older sister) and her (similarly clever and very interesting) friend (from high school) will wait for me.

Clitic third is the position that was never thoroughly analyzed, so I will just give an overview of interesting facts that might be regarded relevant in order to lead the discussions in a new direction.

There are several related phenomena that need to be accounted for when addressing the clitic third pattern. Some of them received unexpected acceptability ratings. I recognize two relevant types of clitic third pattern that have to do with the length and complexity of the first syntactic constituent – one of them is realized when the first syntactic constituent is simple and short and the other is realized when the first syntactic constituent is complex and long.

The first pattern in my data is realized twice with the simple NP and once with an AP followed by a VP which is then followed by a clitic. All of these examples received low ratings, justifying the claims in Croatian grammars.

(11) Eva kupit *će* *mi*
Eva-N buy-Inf will-3-AUX-Cl me-PRON-Dat-Cl
knjigu.
book-Acc
2.1/3.0/2.8
'Eva will buy me a book.'

(12) Jura trebao *ti*
Jura-N supposed you-PRON-Dat-Cl
je sve reći.
be-3-AUX-Pres-Cl all say-Inf
1.8/1.9/2.8
'Jura was supposed to tell you everything.'

(13) Profesoricu srela *sam* jučer.
Professor-Acc met be-1-AUX-Pres-Cl yesterday.
2.6/2.5/2.1[17]
'I met the professor yesterday.'

(14) Sutra donijet *će* *mi*
tomorrow bring-Inf will-3-AUX-Cl me-PRON-Dat-Cl
bilježnicu.

17 Even with the DO fronting, this sentence did not receive significantly lower rating. This might be a sign that when the rating is altogether low, an additional difficulty does not contribute to even lower rating.

notebook-Acc

2.5/3.9/1.8[18]

'He/she will bring me the notebook tomorrow.'

There was only one example of the clitic third pattern after the simple NP (consisting of a sole N) and a VP with an explicit contrastive reading in my data (example (6): Maja kupit *će mi* dar, no ne znam hoće li mi ga darovati 'Maja will buy me a gift, but I am not sure if she is going to give it to me'). The ratings did not increase because of the contrastive reading, and remained in the same range as for the examples without contrast (2.6/1.9/1.8).

If there were no other examples, these sentences would confirm the claims from Croatian grammars. However, numerous counterexamples point the analysis toward a less mechanical and more inclusive theory of grammar. These examples consist of complex first syntactic constituents [NP [NP NP]] followed by a Cl or ClCl.

(15)	Filipova	sestra	i	njezina	prijateljica	Mirta
	Filip's	sister-N	and	her	friend-N	Mirta

odmah	*su*	*ga*	*se*
immediately	be-3-AUX-Cl	him-Gen-Cl	REFL-Cl

sjetile.

remembered

5.8/8.9/5.8

'Filip's sister and her friend Mirta remembered him immediately.'

(16)	Moja	sestra	i	njena	prijateljica	dočekat
	My	sister-N	and	her	friend-N	wait-Inf

će	*nas*	pred	školom.
will-3-AUX-Cl	us-PRON-Acc-Cl	in front of	school-Instr

6.2/8.4/4.1.[19]

18 The only relatively significant variation in this example is seen among the high-school students who rated sentences using ME, but further investigation in this type of clitic third positioning would be needed in order to offer a plausible explanation.

19 One should notice that the elementary school students rated this sentence significantly lower than the high school students. Without a larger body of experimental data it is impossible to suggest the reason for that. Although it might seem plausible to think that the first syntactic constituent in this sentence was too long for the elementary school students to keep track of, they rated two sentences that are of the same length or even longer as being perfectly acceptable, so this is obviously not a plausible explanation.

‘My sister and her friend will wait for us in front of the school.’

Compared to the examples of clitic second after the complex first constituent [$_{NP}$ [NP] [NP]], which are, as already said, claimed to be “a characteristic of sub-standard spoken expression”, but which were also rated as almost perfectly acceptable to all subjects, one realizes that there is no difference in the ratings between these two positions.

Although these are just preliminary results, it has to be said that all other types of clitic third, such as the sentences in which ClCl follow an Adverbial Phrase followed by a VP or the sentences in which Cl or ClCl follow the coordinate NP [$_{NP}$ [N] Conj [N] Conj [N]] followed by an adverb, also receive high ratings, such as in the examples given below. This pattern is extremely frequent on the Internet and in electronic corpora of the Croatian language, which also speaks in favor of its regular use.

(17) Već za nekoliko dana posudit
already in several days lend-Inf
će *mi* bicikl.
will-3-AUX-Cl me-PRON-Dat-Cl bicycle-Acc
6.0/7.2/4.7
‘Already in a few days he/she will lend me (his/her) bicycle.’

(18) Jakov, Miranda i Tvrtko nikad
Jakov-N Miranda-N and Tvrtko-N never
mi neće prestati biti
me-PRON-Dat-Cl won’t-3-AUX-Cl stop-Inf be-Inf
prijatelji.
friends-N
6.5/8.1/6.1
‘Jakov, Miranda and Tvrtko will never stop being my friends.’

One can regard this pattern at least from two angles. If Croatian is considered as a typical Wackernagel type language, which does not favor clitic third as the standard position, the realization of this pattern would elicit a specific IS perspective. From this point of view the sentences which contain clitic third should also be expected to receive at least slightly lower ratings, because the specific IS always calls for limited readings and the rating drops.

Since the given sentences did not receive a rating lower than clitic second sentences, and there is no obvious reason to put the clitic in the third position, I propose a different stance. Given the significant difference in the ratings between examples (11), (12), (13), (14) and examples (16), (17) and (18), one has to take the length of the first syntactic constituent to be the decisive factor. I will formulate the following new rule for Croatian:

> Neutral clitic placement in sentences that do not exhibit a perfect prosodic and syntactic match within the first constituent can be governed by the proximity to the verb, regardless of its position.

The special case

As I already said, there is a special case of Cl and ClCl placement in Croatian. The type of sentences I am talking about are examples in which Cl or ClCl splits the first complex syntactic constituent in such a way that does not satisfy the prosodic or syntactic constraint. The type of sentence presented to the subjects was Moja sestra i njezina *će me* prijateljica dočekati ispred škole. This pattern has been labeled ungrammatical in linguistic literature and was not even mentioned in Croatian grammars, obviously because grammarians treated this pattern to be nonexistent. There is no analysis which can, under a strictly formal view of prosodic and syntactic constraints, produce such a result. In a questionnaire there was one example of this pattern without explicit contrastive reading (19) and two examples with explicit contrastive readings (20 and 21):

(19)	Moja	sestra	i	njen	*me*
	my	sister-N	and	her	me-PRON-Acc-Cl

dečko	slušaju	pažljivo.
boyfriend-N	listen-Pres	carefully

3.3/2.5/3.8

'My sister and her boyfriend are listening to me carefully.'

(20)	Moja	sestra	i	njen	*će*
	my	sister-N	and	her	will-3-AUX-Cl

me	prijatelj,	a	ne	njena	prijateljica,
me-PRON-Acc-Cl	friend-N	and	not	her	friend-N

dočekati	pred	školom.
wait-Inf	in front of	school-Inst

2.8/3.0/2.1

'My sister and her friend, and not her girlfriend, will wait for me in front of the school.'

(21)	Učitelj	Ivan	i	moj	*će*	
	Teacher-N	Ivan-N	and	my	will-3-AUX-Cl	
te		brat,	a	ne	učiteljev,	odvesti
you-PRON-Acc-Cl		brother-N	and	not	teacher's	drive-Inf
na	trening.					
on	practice-Acc					

2.2/1.0/2.1

'The teacher called Ivan and my brother, and not the teacher's (brother) will take you to the practice.'

As can already been expected from all the patterns that I presented until now, both sentences with contrastive reading received lower ratings than the sentence without contrastive reading.[20] The actual puzzle is the rating for sentence (19), whose syntactic structure can be presented as [$_{NP}$ [? [Poss Pron, N, Conj, Poss Pron] Cl [? [N]]], which shows that the complex first syntactic constituent is split in a way so that what is on the left hand side (moja sestra i njen 'my sister and her') is not a constituent and should be treated as a non-splittable island. However, we see that this is not the case and that the rating this utterance received is almost as high as the ratings for the splitting patterns that are claimed to be the prototypical Wackernagel type positions, and significantly higher than the ratings for the clitic third after the simple first constituent pattern.

The results presented in this section are not significantly lower that the results presented for examples (7) and (8) where we see the splitting of the first syntactic constituent after the first prosodic word [$_{NP}$ [N] ClCl [Adj, N]], or than the results of clitic third with the short and simple first syntactic constituent followed by the verb (examples 11, 12, 13 and 14).

I presented the subjects with an additional example of a sentence that should undisputedly be rated as ungrammatical, the source of ungrammaticality lying in the splitting of ClCl. This has been claimed in the literature to be undisputedly ungrammatical. In this sentence separated clitics were placed third and fifth in the sentence, cf.:

(22)	Moji	i	tvoji	prijatelji	doći	*će*
	my	and	your	friends	come-Inf	will-3-AUX-Cl

20 The fact that the first example is rated slightly higher between groups (especially among the high school students who were tested with the use of ME technique, where the difference is already significant) seems to be less important than the fact that several subjects accept these examples and rate them as high as 6 or 7 out of 7, although most of them picked the middle portion of the scale.

sutra	*nam*	u	goste.
tomorrow	us-Dat-Cl	in	guests

‘My and your friends will come to visit us tomorrow.’

It is highly surprising that this example received a fairly high rating (3.0/2.3/3.3), especially when compared to the other already recognized syntactic and prosodic patterns of Cl and ClCl placement. Such a situation requires further investigation, but it is already clear that there is much more variability in Cl and ClCl placement than has been recognized until now, as well as that the interplay of word order and the phonological, syntactic and semantic tiers within the grammatical structure heavily relies on the conceptualization of the entire sentence, as well as on sentence subparts. From this perspective one can find a plausible explanation for the possibility to rate the sentences that violate syntactic rules, but not the prosodic rules within a phonological phrase, such as (19). As concerns example (22), the explanation might be that one should consider the fact that there is a change in progress in Croatian, which allows Cl and ClCl to be placed second in a phonological phrase much further than merely following the first syntactic or prosodic constituent of a sentence. The question that still has to remain open is which conditions allow such placement to be felicitous and whether there are any semantic constraints on the grammaticality of the sentence.

3. Conclusion

In conclusion, I want to stress two points. Firstly, I must emphasize how important it is never to underestimate the complexity and richness of the repertoire of utterances. Secondly, and that is of interest especially for word-order research, it should be clear that there are areas of research in which methodological concerns are at play to a much greater degree than in others. This has to do with the aforementioned complexity and richness of the repertoire. Gradient and limited acceptability in these cases is not necessarily governed (only) by the placement of target elements (such as clitics and clitic clusters), but also by some other, rarely noticed constraints, be that of prosodic, syntactic or semantic kind, or a combination thereof.

Therefore, one should constantly question the empirical status of the research on complex grammatical questions such as word order. From that point of view it is interesting to notice that the ME technique, which has been claimed to offer more fine-grained results in comparison to classic Likert-scale testing, proved to be less sensitive to acceptability judgments, at least in the case of word order.

To sum up, obviously there are several clitic and clitic cluster positions that call for an additional explanation, the most prominent ones being clitic positioning apparently internal to NPs ("splitting") and clitic positioning lower than the second position ("clitic third" or "delayed clitic placement").

It is obvious that, contrary to what has been claimed about word order and clitic placement in Croatian grammars, as well as in the linguistic literature on the matter, we should account for the relative freedom of clitic placement. In contrast to the relative simplicity of data presented in the literature, Cl and ClCl positioning is evidently restricted only by the mere fact that the clitic needs a prosodic constituent to its left on which it can lean.

This gives rise to various segmentations on the sentence level that are governed by the simple prosodic rule that, according to the constraints of prosodic domination, coincides with the phonological phrase (Selkirk 1995). A fair number of speakers obviously adopt certain prosodic (rhythmic) and sentence semantic groupings that govern their preference to allow the acceptance of these highly unexpected syntactic patterns. This is possible only if the examples presented to subjects get acceptable readings on the level of sentence semantics. Although gradient acceptability of sentences with various word order phenomena, including the placement of clitics, heavily relies on the factors that have been unnoticed until now (some of them being rather unexpected, such as the length and the syntactic type of the first (syntactic) constituent), the results presented here show that the that the syntactic functions get to be realized regardless of the placement of clitics in the linear structure, the only change being in the IS of the sentence. This further leads to the conclusion that some speakers might be susceptible to use more resources in realizing (and processing) the details of the information structure than others.

The next step in the research, based on the experimental data I presented here, should be (1) to conduct an experiment with a larger number of participants to confirm the results, (2) to include examples which would control for the complexity of the lexical input, as well as factors detected to be at play in judgments, and (3) to formulate interface rules that would allow for the combinatorial properties in a proper manner.

References

Anderson Stephen R. 1993. "Wackernagel's revenge: Clitics, morphology and the syntax of second position." *Language* 69: 68–98.

Anderson, Steven R. 2006. *Aspects of the Theory of Clitics*. Oxford University Press.

Bard, Ellen Gurman, Dan Robertson, and Antonella Sorace. 1996. “Magnitude estimation of linguistic acceptability.” *Language* 72 (1): 32–68.

Bošković, Željko. 2000. “Second position clitisation: Syntax and/or phonology?” In *Clitic Phenomena in European Languages*, edited by Frits Beukema and Marcel den Dikken, 71–119. Amsterdam: John Benjamins.

Bošković, Željko. 2009. “On Leo Tolstoy, its structure, case, left-branch extraction, and prosodic inversion.” In *A Linguist's Linguist: Studies in South Slavic Linguistics in Honor of E. Wayles Browne*, edited by Steven Franks, Vrinda Chidambaram, and Brian Joseph. 99–122. Bloomington: Slavica.

Browne, Wayles. 1975. “Serbo-Croatian enclitics for English speaking learners.” In *The Zagreb English-Serbo-Croatian project. Contrastive analysis of English and Serbo-Croatian I*, edited by Rudolf Filipović, 105–134. Zagreb: Institute of Linguistics. [Reprinted in Journal of Slavic Linguistics 12 (1–2): 255–289.]

Chafe, Wallace. 1976. “Givenness, contrastiveness, definiteness, subjects, topics, and point of view.” In *Subject and Topic*, edited by Charles Li, 25–55. New York: Academic Press.

Chomsky, Noam, and George A. Miller. 1963. “Introduction to the formal analysis of natural languages.” In *Handbook of Mathematical Psychology*. Vol. 2, edited by R. Duncan Luce, Robert R. Bush, and Eugene Galanter, 269–321. New York: Wiley.

Comrie, Bernard. 1981. *Language Universals and Linguistic Typology*. Chicago: University of Chicago Press.

Cowart, Wayne. 1997. *Experimental Syntax: Applying Objective Methods to Sentence Judgments*. Thousand Oaks: Sage Publications.

Fauconnier, Gilles. 1994. *Mental Spaces: Aspects of Meaning Construction in Natural Language*. Cambridge, New York and Melbourne: Cambridge University Press.

Fodor, Janet D. 2002. “Prosodic disambiguation in silent reading.” *Proceedings of the North East Linguistic Society 32*, edited by M. Hirotani. Amherst: GSLA, University of Massachusetts.

Franks, Steven, and Tracy Holloway King. 2000. *A Handbook of Slavic Clitics*. Oxford University Press.

Halliday, M. A. K. 1967–1968. “Notes on transitivity and theme in English. Parts 1, 2 and 3”. *Journal of Linguistics* 3: 37–81, 199–244; 4: 179–21.

Hofmeister, Philip, Laura Staum Casasanto, and Ivan A. Sag. 2012. “How do individual cognitive differences relate to acceptability judgments? A reply to Sprouse, Wagers, and Phillips.” *Language* 88 (2): 390–400.

Keller, Frank. 2000. “Gradience in grammar: Experimental and computational aspects of degrees of grammaticality.” Ph. D. dissertation, Edinburgh: University of Edinburgh.

Jackendoff, Ray. 2002. *Foundations of Language: Brain, Meaning, Grammar, Evolution*. Oxford University Press.

Jackendoff, Ray. 2007. “Linguistics in cognitive science: The state of art.” *The Linguistic Review* 24: 347–401.

Jackendoff, Ray. 2010. *Meaning and the Lexicon: The Parallel Architecture 1975–2010*. Oxford University Press.

Kaiser, Elsi, and John C. Trueswell. 2004. “The role of discourse context in the processing of a flexible word-order language.” *Cognition* 94(2): 113–147.

Lambrecht, Knud. 1994. *Information Structure and Sentence Form. Topic, Focus, and the Mental Representations of Discourse Referents*. Cambridge University Press.

McClelland, James L. and Joan Bybee. 2007. “Gradience of gradience: A reply to Jackendoff.” *The Linguistic Review* 24: 437–455.

Miller, George A., and Noam Chomsky. 1963. “Finitary models of language users.” In *Handbook of Mathematical Psychology*. Vol. 2. Edited by R. Duncan Luce, Robert R. Bush, and Eugene Galanter, 419–492. New York: Wiley.

Newmayer, Frederick J. 2003. “Grammar is grammar and usage is usage.” *Language* 79 (4): 682–707.

Prince, Ellen F. 1981. “Toward a taxonomy of given/new information.” In *Radical Pragmatics*, edited by Peter Cole, 223–254. New York: Academic Press.

Prince, Ellen F. 1992. “The ZPG letter: Subjects, definiteness, and information-status.” In *Discourse Description: Diverse Analyses of a Fundraising Text*, edited by William C. Mann and Sandra A. Thompson, 295–325. Amsterdam, Philadelphia: John Benjamins.

Progovac, Ljiljana. 2000. “Where do clitics cluster.” In *Clitic Phenomena in European Languages*, edited by Frits Beukema and Marcel den Dikken, 249–258. Amsterdam, Philadlphia: John Benjamins.

Schütze, Carson T. 1996. *The Empirical Base of Linguistics: Grammaticality Judgments and Linguistic Methodology*. Chicago: University of Chicago Press.

Selkirk, Elisabeth. 2001. “The syntax-phonology interface.” In *International Encyclopedia of the Social and Behavioral Sciences*, edited by N. J. Smelser, and Paul B. Baltes, 15407–15412. Oxford: Pergamon.

Sorace, Antonella and Frank Keller. 2005. “Gradience in linguistic data.” *Lingua* 115 (11): 1497–1524.

Sprouse, Jon. 2008. “Magnitude estimation and the non-linearity of acceptability judgments.” *Proceedings of the 27th West Coast Conference on Formal Linguistics*, edited by Natasha Abner, and Jason Bishop, 297–403. Somerville, MA: Cascadilla Proceedings Project.

Staum Casasanto, Laura, Philip Hofmeister, and Ivan A. Sag. 2010. “Understanding acceptability judgments: Additivity and working memory effects.” In *Cognition in Flux: Proceedings of the 32nd Annual Meeting of the Cognitive Science Society*, edited by Stellan Ohlsson, and Richard Catrambone, 224–229. Austin, TX: Cognitive Science Society.

Tomlin, Russel S. 1986. *Basic Word Order: Functional Principles*. London: Croom Helm.

Wasow, Thomas. 2007. “Gradient data and gradient grammars.” *Proceedings from the Annual Meeting of the Chicago Linguistics Society* 43(1): 255–271.